

DeepNoise Project - AI-Assisted Planning and Technical Design

5/27/2026

100 Points Possible

Attempt 1



In Progress

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Unlimited Attempts Allowed

▼ Details

Context

Modern industrial, urban, and technological environments produce large amounts of acoustic information. Machines, motors, alarms, tools, vehicles, and human-made systems generate sounds that can reveal useful information about their current state. For example, a machine may sound normal during regular operation, but its acoustic pattern may change when a component starts to fail, when friction increases, when an impact occurs, or when an alarm is activated.

In this project, your team will design an intelligent acoustic monitoring system based on machine learning and neural networks. The system will receive short audio segments and classify them into different acoustic event categories. For this project, you must define a classification problem with at least **four acoustic classes**. These classes may come from an industrial, environmental, domestic, educational, or technological scenario.

Examples of possible classes include:

1. Normal operation
2. Metallic impact
3. Friction or squeaking
4. Alarm or warning tone

Other examples are also valid, such as animal sounds, environmental sounds, mechanical sounds, simple spoken commands, or

sounds recorded by the team, as long as the classification task is clearly defined and technically feasible.

The main goal of this first deliverable is not to implement the complete system yet. Instead, you will use generative AI tools as design assistants to help you investigate the problem, compare possible solutions, and propose a technically justified plan. However, you must not accept AI-generated answers blindly. You are expected to evaluate, verify, and refine the suggestions provided by AI tools.

Objective

Design a complete technical plan for an acoustic event classification system based on neural networks. The plan must explain how audio data will be collected or selected, how audio signals will be transformed into useful representations, and how a neural network model will be trained and evaluated in the second part of the project.

Specific Objectives

By the end of this first deliverable, your team should be able to:

- Define a realistic acoustic classification problem.
- Select or propose a dataset with at least four classes.
- Explain how raw audio can be transformed into features or image-like representations.
- Compare possible audio representations such as waveform, spectrograms, Mel-spectrograms, and MFCCs.
- Propose a baseline model and a neural network model.
- Justify the use of a CNN for acoustic event classification.
- Design a complete machine learning pipeline.
- Document the use of generative AI tools in a critical and transparent way.

Instructions

Problem Definition

Define the scenario your team will address. You may choose one of the following options or propose a different one:

- Predictive maintenance using machine sounds
- Classification of industrial acoustic events
- Environmental sound classification
- Detection of alarms or warning sounds
- Classification of simple spoken commands
- Classification of sounds recorded by your own team

Your description must include:

- The context of the problem
- Why the classification task is relevant
- The four or more classes that your system will classify
- A short explanation of each class
- Possible challenges, such as noise, class similarity, short recordings, or limited data

Dataset definition

Describe the dataset you plan to use. You may:

- Record your own sounds
- Use a public dataset
- Combine your own recordings with public data

Your dataset proposal must include:

- Source of the data
- Number of classes
- Expected number of audio samples per class
- Approximate duration of each audio sample
- File format
- Sampling rate, if known

- Whether the dataset is balanced or unbalanced
- Possible limitations of the dataset

You may consider public sources such as Freesound, Kaggle, ESC-50, UrbanSound8K, or other open datasets. If you use a public dataset, include the reference or link.

Audio Representation Strategy

Raw audio is a one-dimensional signal, but convolutional neural networks are often applied to two-dimensional structures. Therefore, your team must decide how to represent audio before feeding it into a neural network.

Investigate and compare at least three of the following representations:

- Raw waveform
- STFT spectrogram
- Mel-spectrogram
- MFCCs
- Chroma features
- Log-Mel spectrogram

For each representation, explain:

- What information it captures
- Whether it preserves time and frequency information
- Whether it is suitable for a CNN
- Its advantages and disadvantages
- Whether your team will use it or not

Your team must select one main representation for the project and justify the decision.

A recommended option is the **Mel-spectrogram**, because it converts audio into a two-dimensional time-frequency representation that can be processed by a CNN as an image-like input.

Proposed Machine Learning Pipeline

Design the complete pipeline of your system. Your pipeline should include all major stages from audio acquisition to final prediction.

Your pipeline may follow this structure:

```
Audio files
↓
Preprocessing
↓
Segmentation or duration normalization
↓
Feature extraction or spectrogram generation
↓
Training, validation, and test split
↓
Baseline model
↓
CNN model
↓
Evaluation
↓
Error analysis
```

For each stage, briefly explain what your team will do and why it is necessary.

Baseline Model

Before training a CNN, your team must propose a simple baseline model. The baseline will help you determine whether the CNN actually improves the classification results.

Possible baseline models include:

- k-Nearest Neighbors
- Support Vector Machine
- Random Forest
- Logistic Regression
- Simple MLP

Explain:

- Which baseline model you will use
- What input features it will receive
- Why it is a reasonable comparison
- What result you would expect from this model

Proposed Neural Network

Propose a convolutional neural network architecture for the classification task.

Your proposed CNN does not need to be very deep. A simple architecture is acceptable, for example:

```
Input: Mel-spectrogram
↓
Conv2D + ReLU
↓
MaxPooling2D
↓
Conv2D + ReLU
↓
MaxPooling2D
↓
Flatten
↓
Dense + ReLU
↓
Output layer + Softmax
```

Your description must include:

- Input shape
- Number of output classes
- Type of output activation
- Loss function
- Optimizer
- Evaluation metrics
- Why a CNN is appropriate for this problem

You should also mention possible risks such as overfitting, insufficient data, class imbalance, or poor audio quality.

Validation Strategy

Explain how your team will evaluate the system in the implementation stage.

Your validation plan must include:

- Training, validation, and test split
- Accuracy
- Confusion matrix
- Precision, recall, and F1-score
- Training and validation curves
- Error analysis by class

You should explain why accuracy alone may not be enough, especially if some classes are more difficult to classify or if the dataset is unbalanced.

Critical Use of AI

This project is part of an educational innovation initiative in which AI tools are used to support the solution of a complex problem. Therefore, your team must document how generative AI was used during the planning process.

You may use tools such as ChatGPT, Gemini, Claude, Copilot, or similar systems.

Your report must include an **AI Use Log** with:

- The prompts used
- A short summary of the AI-generated answers
- Which suggestions were useful
- Which suggestions were incorrect, incomplete, or unclear
- How your team verified the technical information
- What decisions were made by the team and not by the AI tool

You must include at least one example where your team critically evaluated an AI-generated suggestion.

Examples of critical reflection questions:

- Did the AI suggest an architecture that was too complex for the dataset?
- Did the AI confuse MFCCs with spectrogram images?
- Did the AI suggest an evaluation metric that was insufficient?
- Did the AI produce code or explanations that required correction?
- What did your team learn by validating the AI-generated information?

Deliverable

Your team must submit a technical planning document that includes:

1. Title of the project
2. Team members
3. Problem definition
4. Description of the acoustic classes
5. Dataset proposal
6. Audio representation comparison
7. Selected representation and justification
8. Complete machine learning pipeline
9. Baseline model proposal
10. CNN architecture proposal
11. Validation strategy
12. AI use log and critical reflection
13. References

Suggested Length

The report should be approximately **4 to 6 pages**, not including references or appendices.

Figures, diagrams, tables, and pipeline illustrations are encouraged.

Submission Specifications

- **Estimated Duration:** 1 week
- **Modality:** Team
- **Submission Format:** PDF report
- **Submission:** Submit via Canvas using the "Submit Assignment" button at the top of this screen.

Evaluation

The report should be approximately **4 to 6 pages**, not including references or appendices.

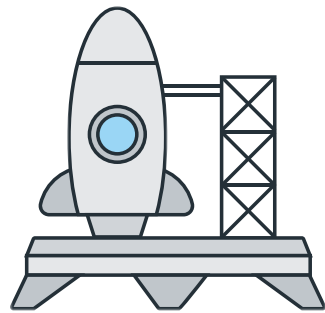
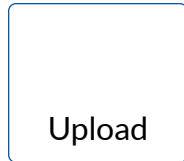
Figures, diagrams, tables, and pipeline illustrations are encouraged.

Evaluation Criteria

Criterion	Description	Points
Problem definition	The acoustic classification problem is clear, relevant, and feasible.	10
Dataset proposal	The dataset is appropriate, clearly described, and compatible with the classification task.	10
Audio representation strategy	The team compares relevant audio representations and justifies the selected one.	10
Machine learning pipeline	The proposed pipeline is complete, coherent, and technically sound.	10
Baseline and CNN proposal	The team proposes a reasonable baseline and a CNN architecture aligned with the problem.	10
Validation strategy	The evaluation plan includes appropriate metrics and error analysis.	10
Critical use of AI	The team documents and critically evaluates the use of generative AI tools.	40
Total		100

Keep in mind, this submission will count for everyone in your Deep Learning Teams group.

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